

Data Quality Report

Generate a comprehensive data quality and profiling report for `model_table.csv` using **ydata-profiling**.

- **Input:** Database/model_table.csv (36 columns, target: `claim_status`)
- **Output:** HTML report + Markdown summary → Notebook PDFs/Phase 2 – Exploratory Data Analysis and Data Quality/

```
In [1]: import os
import sys
import pandas as pd
from ydata_profiling import ProfileReport

/Users/ramsundar/Documents/IITM-Graded-Projects/Capstone/.venv/lib/python3.11/site-packages/tqdm/auto.py:21: TqdmWarning: IPProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
/var/folders/xb/h9lj1q7n4094ljx17mfmchq80000gn/T/ipykernel_10519/2075205170.py:4: DeprecationWarning:
  `import ydata_profiling` is deprecated and will not receive more updates.
  Please install fg-data-profiling via `pip install fg-data-profiling` and use `import data_profiling` instead.

from ydata_profiling import ProfileReport
```

```
In [2]: os.chdir(os.path.join(os.path.dirname(os.getcwd()), ""))
sys.path.append(os.getcwd())
print(f"Working directory: {os.getcwd()}")
```

Working directory: /Users/ramsundar/Documents/IITM-Graded-Projects/Capstone

```
In [3]: df = pd.read_csv("Database/model_table.csv")
print(f"Loaded {len(df)} rows x {len(df.columns)} columns")
df.head()
```

Loaded 25000 rows x 33 columns

Out[3]:

	patient_id	age	gender	city	insurance_provider	chronic_flag	registration_date	visit_id	visit_date	department	...	total_billed	avg_payment_days	provider_rejection_rate	doctor_rejection_rate
0	2964	25	F	Chennai	HealthPlus	1	2025-07-04	3	2025-07-13	ICU	...	44010.80	4.250000	0.149678	0.149678
1	4271	42	F	Mumbai	MediCareX	1	2025-10-31	669	2026-01-17	Neurology	...	180737.86	15.222222	0.152480	0.152480
2	1826	52	M	Pune	CareOne	1	2025-01-23	1264	2025-10-01	General	...	116149.24	9.500000	0.148655	0.148655
3	1224	52	M	Delhi	CareOne	1	2025-06-23	1326	2025-11-05	General	...	93356.58	10.000000	0.148655	0.148655
4	1318	39	M	Chennai	HealthPlus	1	2025-11-13	1455	2025-08-20	Cardiology	...	77804.09	20.600000	0.149678	0.149678

5 rows x 33 columns



```
In [4]: profile = ProfileReport(
        df,
        title="Model Table - Data Quality Report",
        explorative=True,
    )
    print("Profile report generated.")
```

Profile report generated.

```
In [5]: output_dir = "Notebook PDFs/Phase 2 - Exploratory Data Analysis and Data Quality"
        os.makedirs(output_dir, exist_ok=True)

        html_path = os.path.join(output_dir, "data_quality_report.html")
        profile.to_file(html_path)
        print(f"HTML report saved to: {html_path}")
```

```
Summarize dataset: 0%| | 0/5 [00:00<?, ?it/s]
Summarize dataset: 0%| | 0/38 [00:00<?, ?it/s, Describe variable: patient_id]
Summarize dataset: 0%| | 0/38 [00:00<?, ?it/s, Describe variable: gender]
Summarize dataset: 0%| | 0/38 [00:00<?, ?it/s, Describe variable: registration_date]
Summarize dataset: 0%| | 0/38 [00:00<?, ?it/s, Describe variable: department]
Summarize dataset: 3%| | 1/38 [00:00<00:01, 24.98it/s, Describe variable: visit_type]
Summarize dataset: 3%| | 1/38 [00:00<00:01, 23.66it/s, Describe variable: visit_type]
Summarize dataset: 8%| | 3/38 [00:00<00:00, 44.65it/s, Describe variable: risk_score]
Summarize dataset: 11%| | 4/38 [00:00<00:00, 53.00it/s, Describe variable: doctor_id]
Summarize dataset: 13%| | 5/38 [00:00<00:00, 61.87it/s, Describe variable: bill_id]
Summarize dataset: 13%| | 5/38 [00:00<00:00, 61.25it/s, Describe variable: bill_id]
Summarize dataset: 13%| | 5/38 [00:00<00:00, 52.74it/s, Describe variable: bill_id]
Summarize dataset: 13%| | 5/38 [00:00<00:00, 43.78it/s, Describe variable: bill_id]
Summarize dataset: 16%| | 6/38 [00:00<00:00, 51.18it/s, Describe variable: bill_id]
Summarize dataset: 21%| | 8/38 [00:00<00:00, 54.32it/s, Describe variable: bill_id]
Summarize dataset: 24%| | 9/38 [00:00<00:00, 54.64it/s, Describe variable: bill_id]
Summarize dataset: 24%| | 9/38 [00:00<00:00, 54.64it/s, Describe variable: bill_id]
Summarize dataset: 24%| | 9/38 [00:00<00:00, 54.64it/s, Describe variable: billed_amount]
Summarize dataset: 24%| | 9/38 [00:00<00:00, 54.64it/s, Describe variable: approved_amount]
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Summarize dataset: 24%| | 9/38 [00:00<00:00, 54.64it/s, Describe variable: payment_days]
0%| | 0/33 [00:00<?, ?it/s]
Summarize dataset: 32%| | 12/38 [00:00<00:00, 54.64it/s, Describe variable: avg_length_of_stay]
Summarize dataset: 37%| | 14/38 [00:00<00:00, 54.64it/s, Describe variable: total_billed]
Summarize dataset: 37%| | 14/38 [00:00<00:00, 54.64it/s, Describe variable: total_billed]
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Summarize dataset: 39%| | 15/38 [00:00<00:00, 54.64it/s, Describe variable: avg_payment_days]
```


Summarize dataset:	15%		59/401 [00:01<00:09, 36.93it/s, scatter doctor_id, age]
Summarize dataset:	15%		60/401 [00:01<00:09, 36.93it/s, scatter bill_id, age]
Summarize dataset:	15%		61/401 [00:01<00:08, 39.46it/s, scatter bill_id, age]
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Summarize dataset:	18%		71/401 [00:02<00:08, 40.86it/s, scatter visit_month, age]
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Summarize dataset:	32%		127/401 [00:03<00:06, 44.91it/s, scatter visit_month, doctor_id]
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Summarize dataset:	32%		130/401 [00:03<00:06, 44.91it/s, scatter billing_delay_days, doctor_id]
Summarize dataset:	33%		131/401 [00:03<00:05, 45.37it/s, scatter billing_delay_days, doctor_id]
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Summarize dataset:	36%		146/401 [00:03<00:05, 47.12it/s, scatter visit_month, bill_id]
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Summarize dataset:	42%		169/401	[00:04<00:05, 45.86it/s, scatter patient_id, approved_amount]
Summarize dataset:	42%		170/401	[00:04<00:05, 45.86it/s, scatter age, approved_amount]
Summarize dataset:	43%		171/401	[00:04<00:04, 46.46it/s, scatter age, approved_amount]
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Summarize dataset:	44%		176/401	[00:04<00:04, 47.14it/s, scatter billed_amount, approved_amount]
Summarize dataset:	44%		176/401	[00:04<00:04, 47.14it/s, scatter approved_amount, approved_amount]
Summarize dataset:	44%		177/401	[00:04<00:04, 47.14it/s, scatter payment_days, approved_amount]
Summarize dataset:	44%		178/401	[00:04<00:04, 47.14it/s, scatter visit_frequency, approved_amount]
Summarize dataset:	45%		179/401	[00:04<00:04, 47.14it/s, scatter avg_length_of_stay, approved_amount]
Summarize dataset:	45%		180/401	[00:04<00:04, 47.14it/s, scatter days_since_registration, approved_amount]
Summarize dataset:	45%		181/401	[00:04<00:04, 47.49it/s, scatter days_since_registration, approved_amount]
Summarize dataset:	45%		181/401	[00:04<00:04, 47.49it/s, scatter total_billed, approved_amount]
Summarize dataset:	45%		182/401	[00:04<00:04, 47.49it/s, scatter avg_payment_days, approved_amount]
Summarize dataset:	46%		183/401	[00:04<00:04, 47.49it/s, scatter doctor_rejection_rate, approved_amount]
Summarize dataset:	46%		184/401	[00:04<00:04, 47.49it/s, scatter visit_month, approved_amount]
Summarize dataset:	46%		185/401	[00:04<00:04, 47.49it/s, scatter visit_day_of_week, approved_amount]
Summarize dataset:	46%		186/401	[00:04<00:04, 47.77it/s, scatter visit_day_of_week, approved_amount]
Summarize dataset:	46%		186/401	[00:04<00:04, 47.77it/s, scatter days_between_reg_and_visit, approved_amount]
Summarize dataset:	47%		187/401	[00:04<00:04, 47.77it/s, scatter billing_delay_days, approved_amount]
Summarize dataset:	47%		188/401	[00:04<00:04, 47.77it/s, scatter patient_id, payment_days]
Summarize dataset:	47%		189/401	[00:04<00:04, 47.77it/s, scatter age, payment_days]
Summarize dataset:	47%		190/401	[00:04<00:04, 47.77it/s, scatter visit_id, payment_days]
Summarize dataset:	48%		191/401	[00:04<00:04, 47.64it/s, scatter visit_id, payment_days]
Summarize dataset:	48%		191/401	[00:04<00:04, 47.64it/s, scatter length_of_stay_hours, payment_days]
Summarize dataset:	48%		192/401	[00:04<00:04, 47.64it/s, scatter doctor_id, payment_days]
Summarize dataset:	48%		193/401	[00:04<00:04, 47.64it/s, scatter bill_id, payment_days]
Summarize dataset:	48%		194/401	[00:04<00:04, 47.64it/s, scatter billed_amount, payment_days]
Summarize dataset:	49%		195/401	[00:04<00:04, 47.64it/s, scatter approved_amount, payment_days]
Summarize dataset:	49%		196/401	[00:04<00:04, 42.04it/s, scatter approved_amount, payment_days]
Summarize dataset:	49%		196/401	[00:04<00:04, 42.04it/s, scatter payment_days, payment_days]
Summarize dataset:	49%		197/401	[00:04<00:04, 42.04it/s, scatter visit_frequency, payment_days]
Summarize dataset:	49%		198/401	[00:04<00:04, 42.04it/s, scatter avg_length_of_stay, payment_days]
Summarize dataset:	50%		199/401	[00:04<00:04, 42.04it/s, scatter days_since_registration, payment_days]
Summarize dataset:	50%		200/401	[00:04<00:04, 42.04it/s, scatter total_billed, payment_days]
Summarize dataset:	50%		201/401	[00:04<00:04, 43.51it/s, scatter total_billed, payment_days]
Summarize dataset:	50%		201/401	[00:04<00:04, 43.51it/s, scatter avg_payment_days, payment_days]
Summarize dataset:	50%		202/401	[00:04<00:04, 43.51it/s, scatter doctor_rejection_rate, payment_days]
Summarize dataset:	51%		203/401	[00:04<00:04, 43.51it/s, scatter visit_month, payment_days]
Summarize dataset:	51%		204/401	[00:04<00:04, 43.51it/s, scatter visit_day_of_week, payment_days]
Summarize dataset:	51%		205/401	[00:04<00:04, 43.51it/s, scatter days_between_reg_and_visit, payment_days]

Summarize dataset:	51%			206/401	[00:05<00:04, 44.50it/s, scatter days_between_reg_and_visit, payment_days]
Summarize dataset:	51%			206/401	[00:05<00:04, 44.50it/s, scatter billing_delay_days, payment_days]
Summarize dataset:	52%			207/401	[00:05<00:04, 44.50it/s, scatter patient_id, visit_frequency]
Summarize dataset:	52%			208/401	[00:05<00:04, 44.50it/s, scatter age, visit_frequency]
Summarize dataset:	52%			209/401	[00:05<00:04, 44.50it/s, scatter visit_id, visit_frequency]
Summarize dataset:	52%			210/401	[00:05<00:04, 44.50it/s, scatter length_of_stay_hours, visit_frequency]
Summarize dataset:	53%			211/401	[00:05<00:04, 45.08it/s, scatter length_of_stay_hours, visit_frequency]
Summarize dataset:	53%			211/401	[00:05<00:04, 45.08it/s, scatter doctor_id, visit_frequency]
Summarize dataset:	53%			212/401	[00:05<00:04, 45.08it/s, scatter bill_id, visit_frequency]
Summarize dataset:	53%			213/401	[00:05<00:04, 45.08it/s, scatter billed_amount, visit_frequency]
Summarize dataset:	53%			214/401	[00:05<00:04, 45.08it/s, scatter approved_amount, visit_frequency]
Summarize dataset:	54%			215/401	[00:05<00:04, 45.08it/s, scatter payment_days, visit_frequency]
Summarize dataset:	54%			216/401	[00:05<00:04, 45.56it/s, scatter payment_days, visit_frequency]
Summarize dataset:	54%			216/401	[00:05<00:04, 45.56it/s, scatter visit_frequency, visit_frequency]
Summarize dataset:	54%			217/401	[00:05<00:04, 45.56it/s, scatter avg_length_of_stay, visit_frequency]
Summarize dataset:	54%			218/401	[00:05<00:04, 45.56it/s, scatter days_since_registration, visit_frequency]
Summarize dataset:	55%			219/401	[00:05<00:03, 45.56it/s, scatter total_billed, visit_frequency]
Summarize dataset:	55%			220/401	[00:05<00:03, 45.56it/s, scatter avg_payment_days, visit_frequency]
Summarize dataset:	55%			221/401	[00:05<00:03, 45.83it/s, scatter avg_payment_days, visit_frequency]
Summarize dataset:	55%			221/401	[00:05<00:03, 45.83it/s, scatter doctor_rejection_rate, visit_frequency]
Summarize dataset:	55%			222/401	[00:05<00:03, 45.83it/s, scatter visit_month, visit_frequency]
Summarize dataset:	56%			223/401	[00:05<00:03, 45.83it/s, scatter visit_day_of_week, visit_frequency]
Summarize dataset:	56%			224/401	[00:05<00:03, 45.83it/s, scatter days_between_reg_and_visit, visit_frequency]
Summarize dataset:	56%			225/401	[00:05<00:03, 45.83it/s, scatter billing_delay_days, visit_frequency]
Summarize dataset:	56%			226/401	[00:05<00:03, 45.67it/s, scatter billing_delay_days, visit_frequency]
Summarize dataset:	56%			226/401	[00:05<00:03, 45.67it/s, scatter patient_id, avg_length_of_stay]
Summarize dataset:	57%			227/401	[00:05<00:03, 45.67it/s, scatter age, avg_length_of_stay]
Summarize dataset:	57%			228/401	[00:05<00:03, 45.67it/s, scatter visit_id, avg_length_of_stay]
Summarize dataset:	57%			229/401	[00:05<00:03, 45.67it/s, scatter length_of_stay_hours, avg_length_of_stay]
Summarize dataset:	57%			230/401	[00:05<00:03, 45.67it/s, scatter doctor_id, avg_length_of_stay]
Summarize dataset:	58%			231/401	[00:05<00:03, 46.29it/s, scatter doctor_id, avg_length_of_stay]
Summarize dataset:	58%			231/401	[00:05<00:03, 46.29it/s, scatter bill_id, avg_length_of_stay]
Summarize dataset:	58%			232/401	[00:05<00:03, 46.29it/s, scatter billed_amount, avg_length_of_stay]
Summarize dataset:	58%			233/401	[00:05<00:03, 46.29it/s, scatter approved_amount, avg_length_of_stay]
Summarize dataset:	58%			234/401	[00:05<00:03, 46.29it/s, scatter payment_days, avg_length_of_stay]
Summarize dataset:	59%			235/401	[00:05<00:03, 46.29it/s, scatter visit_frequency, avg_length_of_stay]
Summarize dataset:	59%			236/401	[00:05<00:03, 47.00it/s, scatter visit_frequency, avg_length_of_stay]
Summarize dataset:	59%			236/401	[00:05<00:03, 47.00it/s, scatter avg_length_of_stay, avg_length_of_stay]
Summarize dataset:	59%			237/401	[00:05<00:03, 47.00it/s, scatter days_since_registration, avg_length_of_stay]
Summarize dataset:	59%			238/401	[00:05<00:03, 47.00it/s, scatter total_billed, avg_length_of_stay]
Summarize dataset:	60%			239/401	[00:05<00:03, 47.00it/s, scatter avg_payment_days, avg_length_of_stay]
Summarize dataset:	60%			240/401	[00:05<00:03, 47.00it/s, scatter doctor_rejection_rate, avg_length_of_stay]
Summarize dataset:	60%			241/401	[00:05<00:03, 41.36it/s, scatter doctor_rejection_rate, avg_length_of_stay]
Summarize dataset:	60%			241/401	[00:05<00:03, 41.36it/s, scatter visit_month, avg_length_of_stay]

Summarize dataset:	60%		242/401 [00:05<00:03, 41.36it/s, scatter visit_day_of_week, avg_length_of_stay]
Summarize dataset:	61%		243/401 [00:05<00:03, 41.36it/s, scatter days_between_reg_and_visit, avg_length_of_stay]
Summarize dataset:	61%		244/401 [00:05<00:03, 41.36it/s, scatter billing_delay_days, avg_length_of_stay]
Summarize dataset:	61%		245/401 [00:05<00:03, 41.36it/s, scatter patient_id, days_since_registration]
Summarize dataset:	61%		246/401 [00:05<00:03, 42.64it/s, scatter patient_id, days_since_registration]
Summarize dataset:	61%		246/401 [00:05<00:03, 42.64it/s, scatter age, days_since_registration]
Summarize dataset:	62%		247/401 [00:05<00:03, 42.64it/s, scatter visit_id, days_since_registration]
Summarize dataset:	62%		248/401 [00:05<00:03, 42.64it/s, scatter length_of_stay_hours, days_since_registration]
Summarize dataset:	62%		249/401 [00:05<00:03, 42.64it/s, scatter doctor_id, days_since_registration]
Summarize dataset:	62%		250/401 [00:06<00:03, 42.64it/s, scatter bill_id, days_since_registration]
Summarize dataset:	63%		251/401 [00:06<00:03, 43.30it/s, scatter bill_id, days_since_registration]
Summarize dataset:	63%		251/401 [00:06<00:03, 43.30it/s, scatter billed_amount, days_since_registration]
Summarize dataset:	63%		252/401 [00:06<00:03, 43.30it/s, scatter approved_amount, days_since_registration]
Summarize dataset:	63%		253/401 [00:06<00:03, 43.30it/s, scatter payment_days, days_since_registration]
Summarize dataset:	63%		254/401 [00:06<00:03, 43.30it/s, scatter visit_frequency, days_since_registration]
Summarize dataset:	64%		255/401 [00:06<00:03, 43.30it/s, scatter avg_length_of_stay, days_since_registration]
Summarize dataset:	64%		256/401 [00:06<00:03, 43.72it/s, scatter avg_length_of_stay, days_since_registration]
Summarize dataset:	64%		256/401 [00:06<00:03, 43.72it/s, scatter days_since_registration, days_since_registration]
Summarize dataset:	64%		257/401 [00:06<00:03, 43.72it/s, scatter total_billed, days_since_registration]
Summarize dataset:	64%		258/401 [00:06<00:03, 43.72it/s, scatter avg_payment_days, days_since_registration]
Summarize dataset:	65%		259/401 [00:06<00:03, 43.72it/s, scatter doctor_rejection_rate, days_since_registration]
Summarize dataset:	65%		260/401 [00:06<00:03, 43.72it/s, scatter visit_month, days_since_registration]
Summarize dataset:	65%		261/401 [00:06<00:03, 43.77it/s, scatter visit_month, days_since_registration]
Summarize dataset:	65%		261/401 [00:06<00:03, 43.77it/s, scatter visit_day_of_week, days_since_registration]
Summarize dataset:	65%		262/401 [00:06<00:03, 43.77it/s, scatter days_between_reg_and_visit, days_since_registration]
Summarize dataset:	66%		263/401 [00:06<00:03, 43.77it/s, scatter billing_delay_days, days_since_registration]
Summarize dataset:	66%		264/401 [00:06<00:03, 43.77it/s, scatter patient_id, total_billed]
Summarize dataset:	66%		265/401 [00:06<00:03, 43.77it/s, scatter age, total_billed]
Summarize dataset:	66%		266/401 [00:06<00:03, 43.69it/s, scatter age, total_billed]
Summarize dataset:	66%		266/401 [00:06<00:03, 43.69it/s, scatter visit_id, total_billed]
Summarize dataset:	67%		267/401 [00:06<00:03, 43.69it/s, scatter length_of_stay_hours, total_billed]
Summarize dataset:	67%		268/401 [00:06<00:03, 43.69it/s, scatter doctor_id, total_billed]
Summarize dataset:	67%		269/401 [00:06<00:03, 43.69it/s, scatter bill_id, total_billed]
Summarize dataset:	67%		270/401 [00:06<00:02, 43.69it/s, scatter billed_amount, total_billed]
Summarize dataset:	68%		271/401 [00:06<00:02, 43.81it/s, scatter billed_amount, total_billed]
Summarize dataset:	68%		271/401 [00:06<00:02, 43.81it/s, scatter approved_amount, total_billed]
Summarize dataset:	68%		272/401 [00:06<00:02, 43.81it/s, scatter payment_days, total_billed]
Summarize dataset:	68%		273/401 [00:06<00:02, 43.81it/s, scatter visit_frequency, total_billed]
Summarize dataset:	68%		274/401 [00:06<00:02, 43.81it/s, scatter avg_length_of_stay, total_billed]
Summarize dataset:	69%		275/401 [00:06<00:02, 43.81it/s, scatter days_since_registration, total_billed]
Summarize dataset:	69%		276/401 [00:06<00:02, 43.99it/s, scatter days_since_registration, total_billed]
Summarize dataset:	69%		276/401 [00:06<00:02, 43.99it/s, scatter total_billed, total_billed]
Summarize dataset:	69%		277/401 [00:06<00:02, 43.99it/s, scatter avg_payment_days, total_billed]
Summarize dataset:	69%		278/401 [00:06<00:02, 43.99it/s, scatter doctor_rejection_rate, total_billed]

Summarize dataset:	70%		279/401 [00:06<00:02, 43.99it/s, scatter visit_month, total_billed]
Summarize dataset:	70%		280/401 [00:06<00:02, 43.99it/s, scatter visit_day_of_week, total_billed]
Summarize dataset:	70%		281/401 [00:06<00:02, 44.00it/s, scatter visit_day_of_week, total_billed]
Summarize dataset:	70%		281/401 [00:06<00:02, 44.00it/s, scatter days_between_reg_and_visit, total_billed]
Summarize dataset:	70%		282/401 [00:06<00:02, 44.00it/s, scatter billing_delay_days, total_billed]
Summarize dataset:	71%		283/401 [00:06<00:02, 44.00it/s, scatter patient_id, avg_payment_days]
Summarize dataset:	71%		284/401 [00:06<00:02, 44.00it/s, scatter age, avg_payment_days]
Summarize dataset:	71%		285/401 [00:06<00:02, 44.00it/s, scatter visit_id, avg_payment_days]
Summarize dataset:	71%		286/401 [00:06<00:02, 39.40it/s, scatter visit_id, avg_payment_days]
Summarize dataset:	71%		286/401 [00:06<00:02, 39.40it/s, scatter length_of_stay_hours, avg_payment_days]
Summarize dataset:	72%		287/401 [00:06<00:02, 39.40it/s, scatter doctor_id, avg_payment_days]
Summarize dataset:	72%		288/401 [00:06<00:02, 39.40it/s, scatter bill_id, avg_payment_days]
Summarize dataset:	72%		289/401 [00:06<00:02, 39.40it/s, scatter billed_amount, avg_payment_days]
Summarize dataset:	72%		290/401 [00:06<00:02, 39.40it/s, scatter approved_amount, avg_payment_days]
Summarize dataset:	73%		291/401 [00:06<00:02, 41.57it/s, scatter approved_amount, avg_payment_days]
Summarize dataset:	73%		291/401 [00:06<00:02, 41.57it/s, scatter payment_days, avg_payment_days]
Summarize dataset:	73%		292/401 [00:06<00:02, 41.57it/s, scatter visit_frequency, avg_payment_days]
Summarize dataset:	73%		293/401 [00:07<00:02, 41.57it/s, scatter avg_length_of_stay, avg_payment_days]
Summarize dataset:	73%		294/401 [00:07<00:02, 41.57it/s, scatter days_since_registration, avg_payment_days]
Summarize dataset:	74%		295/401 [00:07<00:02, 41.57it/s, scatter total_billed, avg_payment_days]
Summarize dataset:	74%		296/401 [00:07<00:02, 42.99it/s, scatter total_billed, avg_payment_days]
Summarize dataset:	74%		296/401 [00:07<00:02, 42.99it/s, scatter avg_payment_days, avg_payment_days]
Summarize dataset:	74%		297/401 [00:07<00:02, 42.99it/s, scatter doctor_rejection_rate, avg_payment_days]
Summarize dataset:	74%		298/401 [00:07<00:02, 42.99it/s, scatter visit_month, avg_payment_days]
Summarize dataset:	75%		299/401 [00:07<00:02, 42.99it/s, scatter visit_day_of_week, avg_payment_days]
Summarize dataset:	75%		300/401 [00:07<00:02, 42.99it/s, scatter days_between_reg_and_visit, avg_payment_days]
Summarize dataset:	75%		301/401 [00:07<00:02, 44.16it/s, scatter days_between_reg_and_visit, avg_payment_days]
Summarize dataset:	75%		301/401 [00:07<00:02, 44.16it/s, scatter billing_delay_days, avg_payment_days]
Summarize dataset:	75%		302/401 [00:07<00:02, 44.16it/s, scatter patient_id, doctor_rejection_rate]
Summarize dataset:	76%		303/401 [00:07<00:02, 44.16it/s, scatter age, doctor_rejection_rate]
Summarize dataset:	76%		304/401 [00:07<00:02, 44.16it/s, scatter visit_id, doctor_rejection_rate]
Summarize dataset:	76%		305/401 [00:07<00:02, 44.16it/s, scatter length_of_stay_hours, doctor_rejection_rate]
Summarize dataset:	76%		306/401 [00:07<00:02, 44.86it/s, scatter length_of_stay_hours, doctor_rejection_rate]
Summarize dataset:	76%		306/401 [00:07<00:02, 44.86it/s, scatter doctor_id, doctor_rejection_rate]
Summarize dataset:	77%		307/401 [00:07<00:02, 44.86it/s, scatter bill_id, doctor_rejection_rate]
Summarize dataset:	77%		308/401 [00:07<00:02, 44.86it/s, scatter billed_amount, doctor_rejection_rate]
Summarize dataset:	77%		309/401 [00:07<00:02, 44.86it/s, scatter approved_amount, doctor_rejection_rate]
Summarize dataset:	77%		310/401 [00:07<00:02, 44.86it/s, scatter payment_days, doctor_rejection_rate]
Summarize dataset:	78%		311/401 [00:07<00:01, 45.55it/s, scatter payment_days, doctor_rejection_rate]
Summarize dataset:	78%		311/401 [00:07<00:01, 45.55it/s, scatter visit_frequency, doctor_rejection_rate]
Summarize dataset:	78%		312/401 [00:07<00:01, 45.55it/s, scatter avg_length_of_stay, doctor_rejection_rate]
Summarize dataset:	78%		313/401 [00:07<00:01, 45.55it/s, scatter days_since_registration, doctor_rejection_rate]
Summarize dataset:	78%		314/401 [00:07<00:01, 45.55it/s, scatter total_billed, doctor_rejection_rate]
Summarize dataset:	79%		315/401 [00:07<00:01, 45.55it/s, scatter avg_payment_days, doctor_rejection_rate]

Summarize dataset:	79%		316/401 [00:07<00:01, 45.55it/s, scatter avg_payment_days, doctor_rejection_rate]
Summarize dataset:	79%		316/401 [00:07<00:01, 45.55it/s, scatter doctor_rejection_rate, doctor_rejection_rate]
Summarize dataset:	79%		317/401 [00:07<00:01, 45.55it/s, scatter visit_month, doctor_rejection_rate]
Summarize dataset:	79%		318/401 [00:07<00:01, 45.55it/s, scatter visit_day_of_week, doctor_rejection_rate]
Summarize dataset:	80%		319/401 [00:07<00:01, 45.55it/s, scatter days_between_reg_and_visit, doctor_rejection_rate]
Summarize dataset:	80%		320/401 [00:07<00:01, 45.55it/s, scatter billing_delay_days, doctor_rejection_rate]
Summarize dataset:	80%		321/401 [00:07<00:01, 45.71it/s, scatter billing_delay_days, doctor_rejection_rate]
Summarize dataset:	80%		321/401 [00:07<00:01, 45.71it/s, scatter patient_id, visit_month]
Summarize dataset:	80%		322/401 [00:07<00:01, 45.71it/s, scatter age, visit_month]
Summarize dataset:	81%		323/401 [00:07<00:01, 45.71it/s, scatter visit_id, visit_month]
Summarize dataset:	81%		324/401 [00:07<00:01, 45.71it/s, scatter length_of_stay_hours, visit_month]
Summarize dataset:	81%		325/401 [00:07<00:01, 45.71it/s, scatter doctor_id, visit_month]
Summarize dataset:	81%		326/401 [00:07<00:01, 46.39it/s, scatter doctor_id, visit_month]
Summarize dataset:	81%		326/401 [00:07<00:01, 46.39it/s, scatter bill_id, visit_month]
Summarize dataset:	82%		327/401 [00:07<00:01, 46.39it/s, scatter billed_amount, visit_month]
Summarize dataset:	82%		328/401 [00:07<00:01, 46.39it/s, scatter approved_amount, visit_month]
Summarize dataset:	82%		329/401 [00:07<00:01, 46.39it/s, scatter payment_days, visit_month]
Summarize dataset:	82%		330/401 [00:07<00:01, 46.39it/s, scatter visit_frequency, visit_month]
Summarize dataset:	83%		331/401 [00:07<00:01, 47.18it/s, scatter visit_frequency, visit_month]
Summarize dataset:	83%		331/401 [00:07<00:01, 47.18it/s, scatter avg_length_of_stay, visit_month]
Summarize dataset:	83%		332/401 [00:07<00:01, 47.18it/s, scatter days_since_registration, visit_month]
Summarize dataset:	83%		333/401 [00:07<00:01, 47.18it/s, scatter total_billed, visit_month]
Summarize dataset:	83%		334/401 [00:07<00:01, 47.18it/s, scatter avg_payment_days, visit_month]
Summarize dataset:	84%		335/401 [00:07<00:01, 47.18it/s, scatter doctor_rejection_rate, visit_month]
Summarize dataset:	84%		336/401 [00:07<00:01, 41.42it/s, scatter doctor_rejection_rate, visit_month]
Summarize dataset:	84%		336/401 [00:07<00:01, 41.42it/s, scatter visit_month, visit_month]
Summarize dataset:	84%		337/401 [00:07<00:01, 41.42it/s, scatter visit_day_of_week, visit_month]
Summarize dataset:	84%		338/401 [00:08<00:01, 41.42it/s, scatter days_between_reg_and_visit, visit_month]
Summarize dataset:	85%		339/401 [00:08<00:01, 41.42it/s, scatter billing_delay_days, visit_month]
Summarize dataset:	85%		340/401 [00:08<00:01, 41.42it/s, scatter patient_id, visit_day_of_week]
Summarize dataset:	85%		341/401 [00:08<00:01, 42.80it/s, scatter patient_id, visit_day_of_week]
Summarize dataset:	85%		341/401 [00:08<00:01, 42.80it/s, scatter age, visit_day_of_week]
Summarize dataset:	85%		342/401 [00:08<00:01, 42.80it/s, scatter visit_id, visit_day_of_week]
Summarize dataset:	86%		343/401 [00:08<00:01, 42.80it/s, scatter length_of_stay_hours, visit_day_of_week]
Summarize dataset:	86%		344/401 [00:08<00:01, 42.80it/s, scatter doctor_id, visit_day_of_week]
Summarize dataset:	86%		345/401 [00:08<00:01, 42.80it/s, scatter bill_id, visit_day_of_week]
Summarize dataset:	86%		346/401 [00:08<00:01, 43.80it/s, scatter bill_id, visit_day_of_week]
Summarize dataset:	86%		346/401 [00:08<00:01, 43.80it/s, scatter billed_amount, visit_day_of_week]
Summarize dataset:	87%		347/401 [00:08<00:01, 43.80it/s, scatter approved_amount, visit_day_of_week]
Summarize dataset:	87%		348/401 [00:08<00:01, 43.80it/s, scatter payment_days, visit_day_of_week]
Summarize dataset:	87%		349/401 [00:08<00:01, 43.80it/s, scatter visit_frequency, visit_day_of_week]
Summarize dataset:	87%		350/401 [00:08<00:01, 43.80it/s, scatter avg_length_of_stay, visit_day_of_week]
Summarize dataset:	88%		351/401 [00:08<00:01, 44.85it/s, scatter avg_length_of_stay, visit_day_of_week]
Summarize dataset:	88%		351/401 [00:08<00:01, 44.85it/s, scatter days_since_registration, visit_day_of_week]

Summarize dataset:	88%		352/401 [00:08<00:01, 44.85it/s, scatter total_billed, visit_day_of_week]
Summarize dataset:	88%		353/401 [00:08<00:01, 44.85it/s, scatter avg_payment_days, visit_day_of_week]
Summarize dataset:	88%		354/401 [00:08<00:01, 44.85it/s, scatter doctor_rejection_rate, visit_day_of_week]
Summarize dataset:	89%		355/401 [00:08<00:01, 44.85it/s, scatter visit_month, visit_day_of_week]
Summarize dataset:	89%		356/401 [00:08<00:00, 45.27it/s, scatter visit_month, visit_day_of_week]
Summarize dataset:	89%		356/401 [00:08<00:00, 45.27it/s, scatter visit_day_of_week, visit_day_of_week]
Summarize dataset:	89%		357/401 [00:08<00:00, 45.27it/s, scatter days_between_reg_and_visit, visit_day_of_week]
Summarize dataset:	89%		358/401 [00:08<00:00, 45.27it/s, scatter billing_delay_days, visit_day_of_week]
Summarize dataset:	90%		359/401 [00:08<00:00, 45.27it/s, scatter patient_id, days_between_reg_and_visit]
Summarize dataset:	90%		360/401 [00:08<00:00, 45.27it/s, scatter age, days_between_reg_and_visit]
Summarize dataset:	90%		361/401 [00:08<00:00, 45.06it/s, scatter age, days_between_reg_and_visit]
Summarize dataset:	90%		361/401 [00:08<00:00, 45.06it/s, scatter visit_id, days_between_reg_and_visit]
Summarize dataset:	90%		362/401 [00:08<00:00, 45.06it/s, scatter length_of_stay_hours, days_between_reg_and_visit]
Summarize dataset:	91%		363/401 [00:08<00:00, 45.06it/s, scatter doctor_id, days_between_reg_and_visit]
Summarize dataset:	91%		364/401 [00:08<00:00, 45.06it/s, scatter bill_id, days_between_reg_and_visit]
Summarize dataset:	91%		365/401 [00:08<00:00, 45.06it/s, scatter billed_amount, days_between_reg_and_visit]
Summarize dataset:	91%		366/401 [00:08<00:00, 44.50it/s, scatter billed_amount, days_between_reg_and_visit]
Summarize dataset:	91%		366/401 [00:08<00:00, 44.50it/s, scatter approved_amount, days_between_reg_and_visit]
Summarize dataset:	92%		367/401 [00:08<00:00, 44.50it/s, scatter payment_days, days_between_reg_and_visit]
Summarize dataset:	92%		368/401 [00:08<00:00, 44.50it/s, scatter visit_frequency, days_between_reg_and_visit]
Summarize dataset:	92%		369/401 [00:08<00:00, 44.50it/s, scatter avg_length_of_stay, days_between_reg_and_visit]
Summarize dataset:	92%		370/401 [00:08<00:00, 44.50it/s, scatter days_since_registration, days_between_reg_and_visit]
Summarize dataset:	93%		371/401 [00:08<00:00, 44.08it/s, scatter days_since_registration, days_between_reg_and_visit]
Summarize dataset:	93%		371/401 [00:08<00:00, 44.08it/s, scatter total_billed, days_between_reg_and_visit]
Summarize dataset:	93%		372/401 [00:08<00:00, 44.08it/s, scatter avg_payment_days, days_between_reg_and_visit]
Summarize dataset:	93%		373/401 [00:08<00:00, 44.08it/s, scatter doctor_rejection_rate, days_between_reg_and_visit]
Summarize dataset:	93%		374/401 [00:08<00:00, 44.08it/s, scatter visit_month, days_between_reg_and_visit]
Summarize dataset:	94%		375/401 [00:08<00:00, 44.08it/s, scatter visit_day_of_week, days_between_reg_and_visit]
Summarize dataset:	94%		376/401 [00:08<00:00, 38.89it/s, scatter visit_day_of_week, days_between_reg_and_visit]
Summarize dataset:	94%		376/401 [00:08<00:00, 38.89it/s, scatter days_between_reg_and_visit, days_between_reg_and_visit]
Summarize dataset:	94%		377/401 [00:08<00:00, 38.89it/s, scatter billing_delay_days, days_between_reg_and_visit]
Summarize dataset:	94%		378/401 [00:08<00:00, 38.89it/s, scatter patient_id, billing_delay_days]
Summarize dataset:	95%		379/401 [00:08<00:00, 38.89it/s, scatter age, billing_delay_days]
Summarize dataset:	95%		380/401 [00:09<00:00, 38.89it/s, scatter visit_id, billing_delay_days]
Summarize dataset:	95%		381/401 [00:09<00:00, 40.27it/s, scatter visit_id, billing_delay_days]
Summarize dataset:	95%		381/401 [00:09<00:00, 40.27it/s, scatter length_of_stay_hours, billing_delay_days]
Summarize dataset:	95%		382/401 [00:09<00:00, 40.27it/s, scatter doctor_id, billing_delay_days]
Summarize dataset:	96%		383/401 [00:09<00:00, 40.27it/s, scatter bill_id, billing_delay_days]
Summarize dataset:	96%		384/401 [00:09<00:00, 40.27it/s, scatter billed_amount, billing_delay_days]
Summarize dataset:	96%		385/401 [00:09<00:00, 40.27it/s, scatter approved_amount, billing_delay_days]
Summarize dataset:	96%		386/401 [00:09<00:00, 41.42it/s, scatter approved_amount, billing_delay_days]
Summarize dataset:	96%		386/401 [00:09<00:00, 41.42it/s, scatter payment_days, billing_delay_days]
Summarize dataset:	97%		387/401 [00:09<00:00, 41.42it/s, scatter visit_frequency, billing_delay_days]
Summarize dataset:	97%		388/401 [00:09<00:00, 41.42it/s, scatter avg_length_of_stay, billing_delay_days]


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Summarize dataset: 97%|██████████ | 389/401 [00:09<00:00, 41.42it/s, scatter days_since_registration, billing_delay_days]
Summarize dataset: 97%|██████████ | 390/401 [00:09<00:00, 41.42it/s, scatter total_billed, billing_delay_days]
Summarize dataset: 98%|██████████ | 391/401 [00:09<00:00, 42.13it/s, scatter total_billed, billing_delay_days]
Summarize dataset: 98%|██████████ | 391/401 [00:09<00:00, 42.13it/s, scatter avg_payment_days, billing_delay_days]
Summarize dataset: 98%|██████████ | 392/401 [00:09<00:00, 42.13it/s, scatter doctor_rejection_rate, billing_delay_days]
Summarize dataset: 98%|██████████ | 393/401 [00:09<00:00, 42.13it/s, scatter visit_month, billing_delay_days]
Summarize dataset: 98%|██████████ | 394/401 [00:09<00:00, 42.13it/s, scatter visit_day_of_week, billing_delay_days]
Summarize dataset: 99%|██████████ | 395/401 [00:09<00:00, 42.13it/s, scatter days_between_reg_and_visit, billing_delay_days]
Summarize dataset: 99%|██████████ | 396/401 [00:09<00:00, 42.63it/s, scatter days_between_reg_and_visit, billing_delay_days]
Summarize dataset: 99%|██████████ | 396/401 [00:09<00:00, 42.63it/s, scatter billing_delay_days, billing_delay_days]
Summarize dataset: 99%|██████████ | 397/403 [00:09<00:00, 42.63it/s, Missing diagram bar]
Summarize dataset: 99%|██████████ | 398/403 [00:09<00:00, 42.63it/s, Missing diagram matrix]
Summarize dataset: 99%|██████████ | 399/403 [00:09<00:00, 42.63it/s, Take sample]
Summarize dataset: 99%|██████████ | 400/403 [00:09<00:00, 42.63it/s, Detecting duplicates]
Summarize dataset: 100%|██████████ | 401/403 [00:09<00:00, 36.03it/s, Detecting duplicates]
Summarize dataset: 100%|██████████ | 401/403 [00:09<00:00, 36.03it/s, Get alerts]
Summarize dataset: 100%|██████████ | 402/403 [00:09<00:00, 36.03it/s, Get reproduction details]
Summarize dataset: 100%|██████████ | 403/403 [00:09<00:00, 36.03it/s, Completed]
Summarize dataset: 100%|██████████ | 403/403 [00:09<00:00, 42.16it/s, Completed]

Generate report structure: 0%|██████████ | 0/1 [00:00<?, ?it/s]
Generate report structure: 100%|██████████ | 1/1 [00:01<00:00, 1.76s/it]
Generate report structure: 100%|██████████ | 1/1 [00:01<00:00, 1.76s/it]

Render HTML: 0%|██████████ | 0/1 [00:00<?, ?it/s]
Render HTML: 100%|██████████ | 1/1 [00:00<00:00, 2.27it/s]
Render HTML: 100%|██████████ | 1/1 [00:00<00:00, 2.27it/s]

Export report to file: 0%|██████████ | 0/1 [00:00<?, ?it/s]
Export report to file: 100%|██████████ | 1/1 [00:00<00:00, 86.49it/s]
HTML report saved to: Notebook PDFs/Phase 2 – Exploratory Data Analysis and Data Quality/data_quality_report.html
```